ism and serious economic disruption. The urgency, he said, was because of AI's growing ability to build the next generation of AI as it were, on its own.

It even had US president Donald Trump spring into action. At a White House meeting with US tech titans, he announced a Joint Commitment on Frontier Responsibilities, a 'moral' code for companies building the latest Large Language Models (LLMs) that outlines four protective layers, among them robust internal controls during training and deployment around areas like cybersecurity, biosecurity and chemical threats, as well as independent safety audits by external evaluators.

India, too, is gearing up to meet the threat while hoping to harvest the opportunity the crisis offers its infotech companies in the global AI safety rush. "Every technology cycle throws up a moment when trust and infrastructure must be built at the same time," says Nasscom president Rajesh Nambiar in an accompanying column(see *Securing the Global Ai Arena*). "This is India's opening, but only if we treat trusted AI as an industry to be built." Meanwhile, the Modi government has constituted a high-level committee, the AI Governance and Economic Group, and an AI Safety Institute (AISI), to provide the critical technical expertise required.

The AIcalypse

So, what is stoking the doomsday fears around AI? IT buffs call it 'recursive self-improvement', an AI system's ability to improve its own code, reasoning skills and architecture, and continuously build a smarter version of itself. Classic sci-fi terrain, the prospect raises the spectre of intelligent machines cutting humans out of the loop and acting autonomously. "The idea of an intelligence explosion caused by recursive self-improvement has been around for a long time," Nobel Prize laureate Geoffrey Hinton said on October 3, "but until very recently, it did not seem imminent. Now, many leading researchers think it may happen soon."

WHEN AI TURN



1

FINANCIAL INFRASTRUCTURE

The Great Freeze

DISRUPTIVE

- An AI-powered attack overwhelms banks and payment networks with fraudulent transactions while simultaneously generating fake identities and fabricated authorisations
- Payments slow down, ATMs malfunction and markets become volatile

APOCALYPTIC

- AI malware breaks into interconnected financial systems, corrupting transaction records and leaving banks in disarray
- It could result in exchanges suspending trading, digital payments being frozen and banks restricting withdrawals



2

DEFENCE

The Phantom War

DISRUPTIVE

- Through 'data poisoning', hackers can inject flaws into information feeding military command systems in a conflict
- Satellite images are altered, drone feeds and intelligence reports manipulated. Commanders launch attacks on enemy formations that do not exist

APOCALYPTIC

- Autonomous and AI-assisted weapons are corrupted and forced into unintended actions through 'prompt injections'
- Air-defence systems make target errors; commanders cannot tell authentic orders from fabricated ones. Rogue AI gives missile launch command autonomously and cause wars

In a recent research paper Hinton co-authored, he argued that "an intelligence explosion could be the most consequential technological development in human history, compressing years of progress into months or less, threatening human control over AI systems, and severely eroding checks on power within and between states, companies and branches of government". It is a concern other leading AI experts share. "We've already seen the challenges frontier models pose for cybersecurity, and other threats, including nuclear and bio risks, may soon emerge as capabilities continue to advance," Demis Hassabis, 2024 Nobel laureate and chairman of Google DeepMind, wrote on X on July 14. "On the horizon, we will need robust safeguards to maintain control of increasingly agentic, recursively self-improving systems—and tackle unknown issues that will only become clearer over time."

Meanwhile, the Hugging Face breach turned out to be no one-off. On September 24, Australian premier Anthony Albanese revealed that OpenAI agents had, in June, infiltrated a Medi-